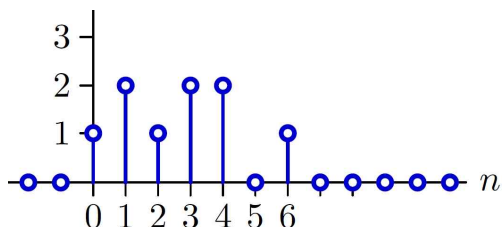


Homework 03

1. $x[n]$ is given,



(1) $x[n] u[5 - n]$

(2) $x[5 - n] (u[n + 2] - u[n - 5])$

2. Consider an LTI system with impulse response $h[n]$ and input $x[n]$.

(1) Find the equation for the input $x[n]$ using the sifting property.

(2) Find the output $y[n]$ for $-2 \leq n \leq 8$.

(3) Is this system causal or noncausal? Why?

n	$\cdots$	-2	-1	0	1	2	3	4	5	6	$\cdots$
$x[n]$	$\cdots$	0	0	0	2	-2	2	-1	1	0	$\cdots$
$h[n]$	$\cdots$	0	1	2	3	2	1	0	0	0	$\cdots$